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Learning chemical-induced gene expression perturbations with WAVE

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Chemical-induced transcriptional profiles facilitate systematic characterization of compound-driven perturbations, offering critical insights into mechanisms of action, especially for compounds with poorly defined molecular targets. However, experimentally profiling gene expression across all combinations of cell lines and compounds is impractical, limiting the large-scale application of comparative transcriptomics in drug screening. To address this, we developed a deep learning framework named WAVE (Wave Action-of-Drug with Variational Encoder), which predicts gene expression profiles from chemical structures and the basal states of cell lines and single cells. By effectively integrating drug molecular representations and cellular basal transcriptional profile using a β-variational autoencoder (β-VAE) framework, WAVE accurately predicts untested chemical-induced transcriptional profiles. Leveraging this integration of cellular insights and molecular characteristics, WAVE excels in drug discovery and drug repurposing based on in silico perturbation experiment. It's application in lung adenocarcinoma (LUAD) drug discovery showcased its potential to identify potentially effective compounds. Our results demonstrate the utility of WAVE for elucidating mechanisms of action for potential drugs and enabling large-scale drug screening.

Phenotype-based drug screening is an important drug discovery strategy that directly identifies compounds with potential therapeutic effects by observing their impact on the phenotypic characteristics of organisms, without relying on molecular target information[1]. However, this approach struggles to reveal the molecular mechanisms of drugs. In recent years, comparative transcriptomics-based strategies have increasingly become vital tools for studying drug mechanisms. By analyzing changes in gene expression before and after compound treatment, researchers can gain deeper insights into a drug's effects on key genes, signaling pathways, and gene regulatory networks, uncovering the potential biological efficacy of compounds[2]. Compared to traditional phenotype-based approaches, transcriptomics-based drug discovery strategies offer deeper insights into the mechanisms of drug action, potentially enhancing the precision and efficiency of drug screening, particularly for compounds with unknown or poorly characterized targets[3].

With advancements in high-throughput sequencing technology, a series of large-scale public gene expression databases based on drug-

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treated cell lines have been established, providing crucial data for drug screening and discovery, including CMap[4], GDSC[5], CCLE[6], and LINCS L1000[7]. The CMap database comprises approximately 1,300 drug-induced transcriptional profiles across five human cancer cell lines. Its successor, LINCS, significantly expands this resource by profiling ~20,000 compounds across ~180 human cell lines, resulting in over 1.4 million gene expression signatures. Despite the abundance of transcriptional data on compound-induced transcriptional profiles in the LINCS L1000 dataset, a vast number of compound–cell line perturbation combinations remain uncharacterized. This constraint severely restricts the exploration of new perturbation conditions involving different compounds and cell lines.

To address this limitation, recent efforts have turned to computational strategies that aim to predict unseen perturbation signatures. These approaches typically involve learning compound–cell interactions from existing drug-induced gene expression profiles and generalizing to unseen compound–cell line pairs, thereby enabling in silico expansion of perturbation space and facilitating more comprehensive drug discovery. DeepCOP[8] is a deep learning framework designed to predict gene regulatory effects of small molecules by integrating gene ontology (GO) annotations, molecular fingerprints, and cell line descriptors. DLEPS[9] employs a variational autoencoder (VAE) model to predict gene expression changes induced by compounds, though its predictions lack cell type specificity. CIGER[10] and DeepCE[11] utilize graph convolutional networks and multi-head attention mechanisms, integrating molecular fingerprints, cell line features, and gene expression to predict perturbation profiles. MultiDCP[12] incorporates prior knowledge from STRING[13] and CCLE databases, combining molecular graph networks and baseline cell line expression to predict post-perturbation profiles.

Among these approaches, generative models have emerged as a particularly promising direction for gene expression profile prediction. CPA[14] uses an autoencoder-based model to map chemically induced transcriptomic effects into a latent space for perturbation response reconstruction. ChemCPA[15] introduces an innovative encoder-decoder framework integrating compound chemical structures to predict unknown drug perturbation effects. TranSiGen[16] trains separate VAE models on pre- and post-perturbation expression profiles, requiring only baseline expression and simplified molecular input line entry system (SMILES) strings for prediction. Recent large-scale perturbation modeling efforts have further highlighted the importance of generalizing perturbation effects across diverse biological contexts, particularly in single-cell settings, using context-aware representation learning frameworks[17]. Deep generative models have greatly advanced various fields, demonstrating significant potential in drug discovery[18].

Here, we propose WAVE, a deep generative model-based method for predicting compound-induced gene expression profiles at both bulk and single-cell levels. WAVE is built on a β-VAE encoder-decoder architecture, consisting of three components: gene variational autoencoder, compound feature network and gene-compound fusion module. Unlike standard VAE, β-VAE explicitly controls the trade-off between reconstruction fidelity and latent regularization through the β coefficient, encouraging more structured and disentangled latent representations[19]. We adopted this design because such regularization is expected to improve the robustness of the latent space and reduce overfitting to cell-specific transcriptional patterns, which is particularly important for perturbation prediction under unseen biological contexts. For molecular representation, we adopted Morgan fingerprints, a computationally efficient circular fingerprinting scheme that remains highly competitive against graph-based molecular representations in a variety of molecular analysis tasks[20,21].

WAVE takes compound SMILES strings and unperturbed cell expression profiles as input to predict perturbed gene expression, making it applicable to untested compounds and disease perturbations across various scenarios. It learns the latent space of gene expression and chemical interactions, inferring cell-type-specific differential gene expression profiles. Benchmark results show that WAVE outperforms other methods in predicting untested compounds and cell lines. WAVE offers a scalable framework for modeling transcriptional responses to chemical perturbations. To explore its utility in a clinically relevant context, we applied it to identify candidate compounds with therapeutic activity against LUAD (lung adenocarcinoma). Its flexibility and accuracy make it a powerful auxiliary tool for candidate drug discovery. Furthermore, we have developed a robust computational pipeline based on WAVE to identify candidate drugs with potential efficacy against specific diseases. By integrating this pipeline with gene pathway analysis, WAVE can be used to evaluate the potential therapeutic efficacy of compounds for specific diseases.

Results

Overview of the WAVE framework

The transcriptional profile change reflects alterations in the transcriptional state of cells induced by covariates such as compound structure, dosage, and exposure duration. WAVE adopts the experimental conditions of 10 μM compound concentration and 24 h treatment duration to focus the model on the impact of compound structure on transcriptional profiles. WAVE aims to generalize from known compound–cell combinations to unseen ones. Morgan molecular fingerprints were employed to encode the chemical structure of compounds, while a β-VAE encoder was used to extract latent representations from gene expression profiles. The lower-dimensional embeddings of gene expression and drug fingerprints are concatenated and subsequently processed by a series of fully connected layers, designed to capture the differential patterns of gene expression induced by drug perturbation. The latent representation of gene expression is then input into the β-VAE decoder to obtain denoised, unperturbed gene expression. The differential changes in gene expression are added to this reconstructed, denoised, unperturbed gene expression to ultimately generate the post-perturbed gene expression profile.

WAVE is an extensible perturbation-conditioned generative model that can be trained on both bulk and single-cell transcriptomic data to predict transcriptional responses to untested chemical perturbations (Fig. 1a). WAVE consists of three main components: a gene variational autoencoder, a molecular feature network, and a gene-compound fusion module (Fig. 1b).

The Gene Variational Autoencoder learns latent representations of gene expression data. This design enables the model to capture the underlying structure and variability of gene expression patterns, providing a compact and informative representation that can be used for downstream prediction tasks.

The Molecular Feature Network is designed to extract informative representations of chemicals by transforming molecular fingerprints into a compact latent space. By leveraging this low-dimensional embedding, the model can capture key structural and functional properties of compounds in a way that facilitates efficient integration with gene expression data.

The Gene-Compound Fusion Module integrates the latent representation of gene expression with the molecular embedding of the compound by concatenating them and feeding the combined vector into a series of fully connected layers. This design enables the model to capture complex nonlinear interactions between cellular states and chemical perturbations, ultimately generating the transcriptional response induced by the drug.

Finally, the transcriptional response changes are added to the reconstructed baseline gene expression from the β-VAE decoder to obtain the predicted post-perturbed gene expression profile.

To better understand the distribution and characteristics of the training data used in WAVE, we analyzed the composition of the LINCS

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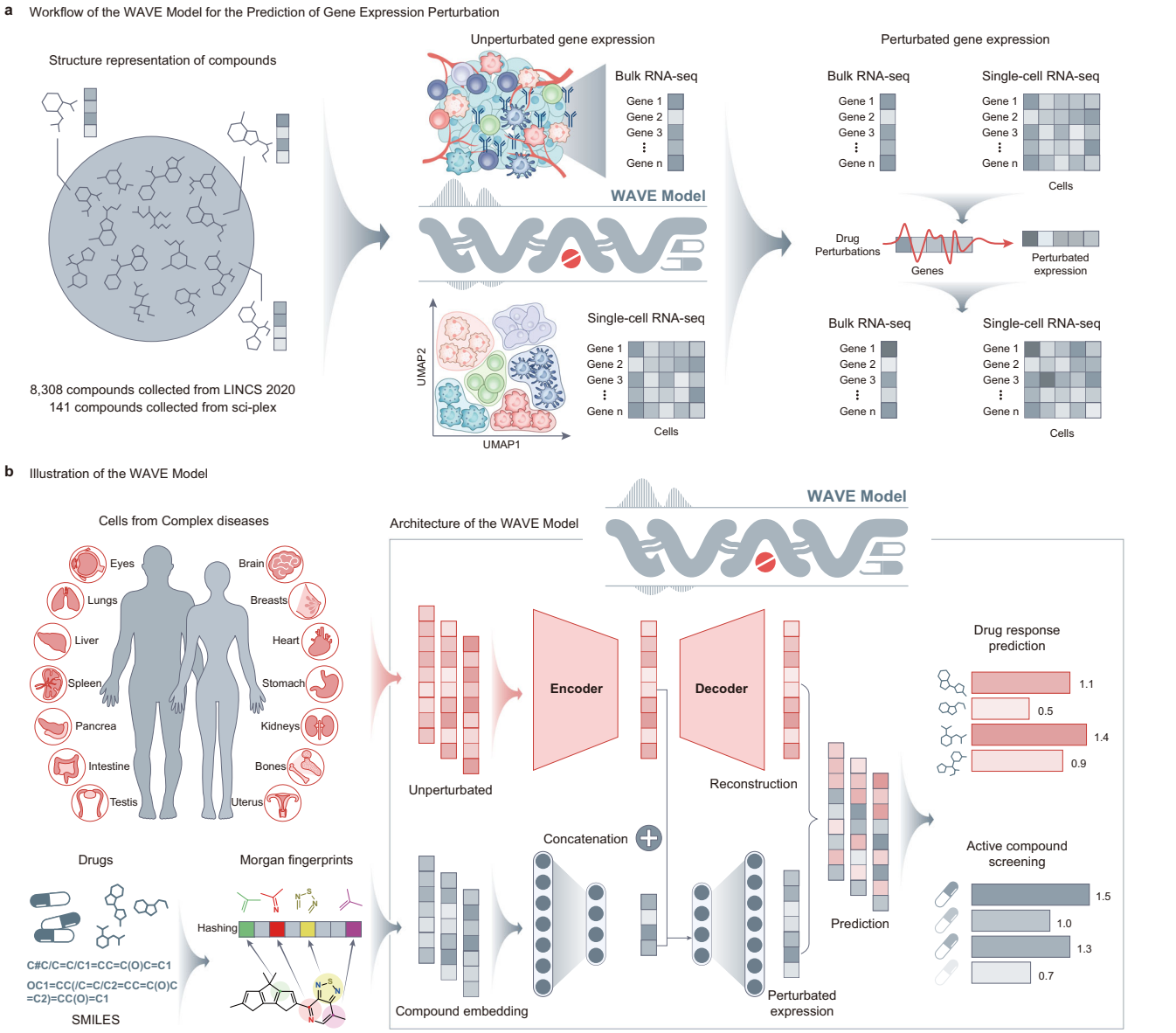


Fig. 1 | Overview of the WAVE model for compound-induced gene expression prediction. a The WAVE framework integrates chemical structure information with transcriptomic profiles to predict drug-induced gene expression perturbations. Structural representations from the LINCS 2020 dataset and sci-Plex dataset are encoded and combined with unperturbed gene expression profiles to learn cell type-specific transcriptional responses. **b** WAVE adopts an encoder–decoder framework that integrates unperturbed gene expression and compound structure embeddings to predict drug-perturbed transcriptional profiles. The model concatenates cell type-specific expression features with molecular fingerprints derived from SMILES, enabling reconstruction of compound-specific responses. This architecture supports downstream tasks such as drug response prediction and compound prioritization.

and sci-Plex datasets. In the LINCS dataset, drug perturbations span a diverse range of cell lines derived from various tissues and organs, with compounds targeting a wide spectrum of proteins and mechanisms of action (MoAs). The sci-Plex dataset provides single-cell resolution perturbation profiles across three major cell lines (MCF7, A549, and K562), covering a broad array of pathways and drug targets. A detailed summary of cell line origins, top drug targets, MoAs, and associated pathways is provided in Supplementary Fig. 1, offering a comprehensive overview of the pharmacological and biological diversity present in both datasets.

Accurately representing molecular structures is crucial for predicting compound effects. To achieve this, WAVE employs Morgan fingerprints, which are widely used in cheminformatics for encoding molecular information. At the bulk level, the training data are sourced from the LINCS 2020 Level 3 dataset, which collects gene expression before and after compound perturbations for 978 landmark genes. Only drug-treated samples under the screening condition of 24 h at 10 μM were retained, with dimethyl sulfoxide (DMSO) treated groups serving as controls. For data within the same group, the MODZ method was applied for data integration. Supplementary Table 1 summarizes the hyperparameters of the WAVE model.

WAVE accurately predicts compound-induced transcriptional profile changes

We split the LINCS 2020 dataset into training and test sets at an 8:2 ratio and employed 5-fold cross-validation for training. Prediction performance on the test set was evaluated using the coefficient of determination (R^2) and Pearson correlation coefficient (PCC) as metrics. Both R^2 and PCC were found to be close to 1, with the area under the curve (AUC) reaching 0.938 (Fig. 2a) and 0.969 (Fig. 2b), respectively, confirming that WAVE can accurately predict post-perturbation gene expression profiles.

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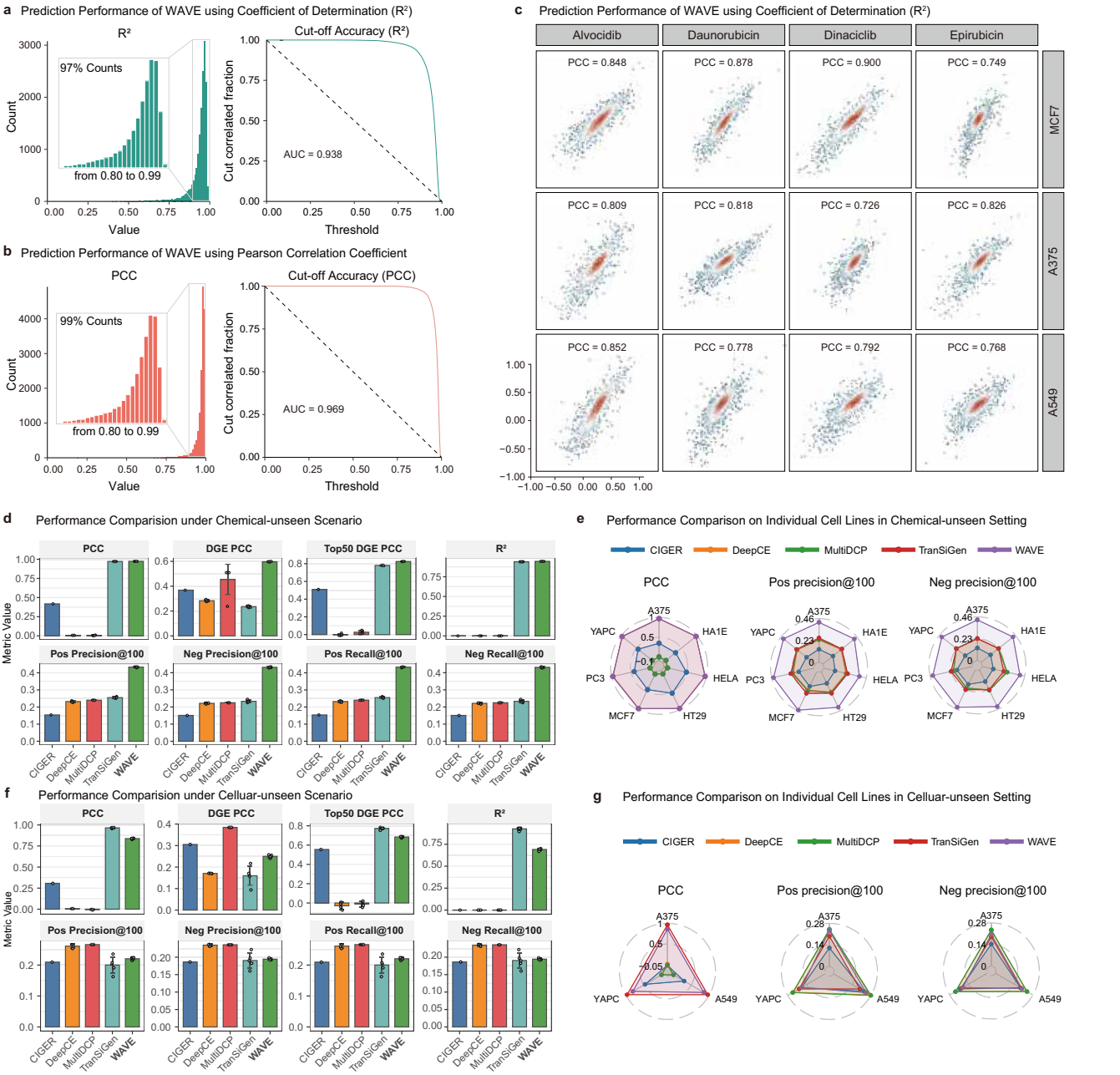


Fig. 2 | Evaluation of WAVE performance in compound-induced gene expression prediction. a Distribution of R^2 across all test samples (left) and the area under the curve (AUC) = 0.938 (right). **b** Distribution of PCC between predicted and true profiles (left) and cumulative accuracy curve (right), with AUC = 0.969. **c** Representative scatter plots of predicted vs. true log2FC values for 4 anticancer agents (alvocidib, daunorubicin, dinaciclib, and epirubicin) in 3 main LINCS cell lines (MCF7, A375, and A549), with corresponding PCC values. **d** Performance comparison of five models (CIGER, DeepCE, MultiDCP, TranSiGen, and WAVE) under the chemical-unseen scenario, evaluated by PCC, DGE PCC, and top 50 DGE PCC. Bars show the mean performance across five cross-validation folds; error bars indicate s.d. across folds ($n = 5$). **e** Radar plots showing model performance on individual cell lines in the chemical-unseen setting. **f** Performance comparison of the five models under the cellular-unseen scenario, evaluated by PCC, DGE PCC, top 50 DGE PCC, R^2, positive precision@100, negative precision@100, positive recall@100, and negative recall@100. Bars show the mean performance across five cross-validation folds; error bars indicate s.d. across folds ($n = 5$). **g** Radar plots showing model performance on individual cell lines in the cellular-unseen setting.

Differential gene expression (DGE) analysis in response to drug perturbation is essential for understanding compound MoAs. Since accurate prediction of drug-induced differential expression is particularly important for understanding compound MoAs, we further assessed WAVE using perturbation-focused metrics, including DGE PCC, Top-50 DGE PCC, positive/negative precision@100, and positive/negative recall@100. Across the three evaluation settings, including the random-split, chemical-unseen, and cellular-unseen scenarios, WAVE consistently showed strong performance on these complementary metrics (Supplementary Fig. 2a). In particular, the DGE PCC reached approximately 0.29 in the cellular-unseen setting and around 0.57 and 0.60 in the chemical-unseen and random settings, respectively, with similarly robust performance observed for Top-50 DGE PCC and gene prioritization metrics.

To assess the performance of WAVE in capturing $\log_2$ fold change ($\log_2$FC), we collected prediction results for two CDK inhibitors (alvocidib, dinaciclib) and two anthracycline antibiotics (daunorubicin, epirubicin) across three major LINCS cell lines (MCF7, A375, A549) and

compared them with the true log_2FC values. The predicted log_2FC values achieved PCCs greater than 0.74 across all compound–cell line combinations (Fig. 2c).

To systematically assess the generalization capability of the model, we designed two benchmark scenarios: chemical-unseen and cellular-unseen, which simulate the prediction of transcriptional responses for unseen compounds and unseen cell lines, respectively. To evaluate model performance, we adopted three metrics: PCC, DGE PCC, and top-50 DGE PCC. PCC measures the overall consistency between the predicted and ground-truth gene expression profiles. DGE PCC evaluates the consistency between predicted DGE and true DGE. Since the top up- and down-regulated genes often drive the most pronounced biological effects, we further used Top-50 DGE PCC, positive/negative precision@100, and positive/negative recall@100 to assess the biological relevance of the predictions.

In the chemical-unseen scenario, we constructed a benchmark dataset by collecting 27,095 compound–cell line pairs, encompassing 7,483 unique compounds. Since CIGER, DeepCE, and MultiDCP were specifically trained on the seven major LINCS cell lines (A375, HA1E, HELA, HT29, MCF7, PC3, and YAPC), we performed comparative evaluations using data from these cell lines. The dataset was split into training and testing sets at a ratio of 8:2, ensuring that a portion of the compounds remained unseen during training. 5-fold cross validation was applied to train the models. Under this setting, WAVE achieved the best overall performance across multiple evaluation metrics (Fig. 2d). In terms of global profile similarity, WAVE achieved the highest PCC, approaching 1 (Fig. 2d). While TranSiGen also reached a PCC close to 1, it underperformed compared to WAVE in terms of DGE PCC and Top-50 DGE PCC (Fig. 2d). WAVE also showed clear advantages in both precision@100 and recall@100 for up-regulated and down-regulated genes (Fig. 2d). In addition, WAVE demonstrated consistently high prediction accuracy across all individual cell lines (Fig. 2e).

In the cellular-unseen setting, we collected 37,105 compound–cell line pairs. The model was trained on data from 6 cell lines (HA1E, HELA, HT29, MCF7, PC3, and VCAP) and tested on 3 other cell lines (A375, YAPC, and A549) to assess generalization performance. Although some baseline models achieved higher scores on one or two individual metrics, these gains were often accompanied by clear weaknesses in other key metrics. For instance, TranSiGen showed the highest PCC and strong R^2, but its precision@100 and recall@100 values were consistently lower than those of several other methods, suggesting limited ability to prioritize the most relevant differentially expressed genes (Fig. 2f). In contrast, DeepCE and MultiDCP achieved relatively strong precision/recall metrics, but their much lower PCC, Top-50 DGE PCC, and R^2 indicated weaker recovery of global and perturbation-specific transcriptional patterns (Fig. 2f). WAVE, by comparison, maintained competitive or leading performance across nearly all major metrics, including PCC, DGE PCC, Top-50 DGE PCC, R^2, and both positive and negative precision/recall@100 (Fig. 2f). This balanced performance highlights its superior robustness and generalizability to unseen cellular contexts. Figure 2g provides a detailed breakdown of model performance across individual test cell lines.

WAVE captures cell lineage-specific transcriptional perturbations across the LINCS dataset

To explore whether WAVE can capture lineage-specific transcriptional perturbations, we visualized the cell embeddings from the test set using uniform manifold approximation and projection (UMAP). The result revealed that WAVE effectively learns gene expression patterns across cell lines and preserves their biological heterogeneity (Fig. 3a). Meanwhile, for cell lines derived from the same tissue, the embeddings learned by WAVE exhibited closer proximity in UMAP space (e.g., breast and lung cell clusters) (Fig. 3a, Supplementary Fig. 2b). This indicates that WAVE can effectively preserve tissue-specific expression heterogeneity, demonstrating its potential to uncover transcriptional response variations.

We next compared the predicted and true expression embeddings in a side-by-side manner. To ensure robust visualization, we selected only cell lines that had more than 500 compound perturbation profiles available for analysis. As shown in Fig. 3b (left panel), the predicted profiles largely overlapped with the ground truth profiles in UMAP space, demonstrating that WAVE achieves accurate global reconstruction of the transcriptional landscape. Furthermore, when coloring cells by their corresponding cell lines (Fig. 3b, right panel), distinct clusters emerged for each cell line, reflecting that WAVE not only reconstructs the overall expression patterns but also captures fine-grained cell line-specific transcriptional signatures. To explore the broader applicability of WAVE, we used the model to predict transcriptomic responses for 115,232 previously uncharacterized compound–cell line pairs in the LINCS dataset. Visualization by UMAP method of the predicted post-perturbation gene expression revealed well-separated clusters, indicating that WAVE captures meaningful biological variation across different drug perturbations and cell types (Fig. 3c). Notably, clustering patterns aligned with known tissue origins (e.g., lung, blood, breast), suggesting that the predicted responses retain cell-type-specific transcriptional signatures.

These results demonstrate that WAVE not only predicts global transcriptional patterns but also prioritizes the most biologically relevant gene-level changes across diverse cellular contexts. To assess WAVE's ability to generalize transcriptional responses across compounds and cell types, we evaluated the hold-out perturbation profiles of entinostat, a class I HDAC inhibitor with anticancer activity. We evaluated whether WAVE could accurately predict its transcriptional effects across diverse cell types. As shown in the differential expression gene (DEG) heatmap across multiple cell lines (Fig. 3d), WAVE robustly recapitulated the hallmark transcriptional effects of HDAC inhibition, including consistent upregulation of genes such as, *BRCA1*, which is associated with cell cycle arrest and immune modulation[22–24]. Conversely, down-regulation of metastasis- and invasion-related genes such as *MMP1*, *TIAM1*, and *FYN* supports entinostat's anti-invasive properties[25–27]. This pattern not only aligns with known HDAC mechanisms but also demonstrates WAVE's ability to model nuanced transcriptional changes across diverse cancer types.

To investigate the biological implications of entinostat-induced transcriptional changes, we expanded the predicted gene expression from 978 landmark genes to 11,832 genes using the LINCS imputation matrix[7]. Then, we conducted gene set enrichment analysis (GSEA) to underlie the signal pathway impact across cell types. The dot plot of biological process enrichment (Supplementary Fig. 2c) revealed that entinostat activates gene programs related to mitochondrial translation, cholesterol and sterol biosynthesis, and autophagy, all of which have been implicated in HDAC inhibitor-mediated anti-cancer mechanisms. In parallel, ribosome biogenesis and DNA repair-related pathways were found to be strongly suppressed, aligning with reduced proliferation and genome maintenance capacity in cancer cells.

To further assess the generalizability of WAVE, we explored WAVE's ability to predict pathway alterations induced by other compounds with known targets. We performed GSEA analysis for drugs targeting specific pathways, including ACE, BRAF, and NR3C1. The results of these analyses are shown in Supplementary Fig. 2d–f, which provide insights into the pathway-level effects of each drug. For ACE inhibitors, the enriched pathways primarily involved glycolytic metabolic processes, cell adhesion, and immune modulation. BRAF inhibitors, on the other hand, activated pathways related to DNA replication, chromosome organization, and cell cycle regulation. In contrast, NR3C1 inhibition predominantly affected pathways involved in RNA splicing, DNA repair, and angiogenesis. These findings, combined with the GSEA enrichment scores, highlight WAVE's robust capability in predicting transcriptional shifts across different drug perturbations

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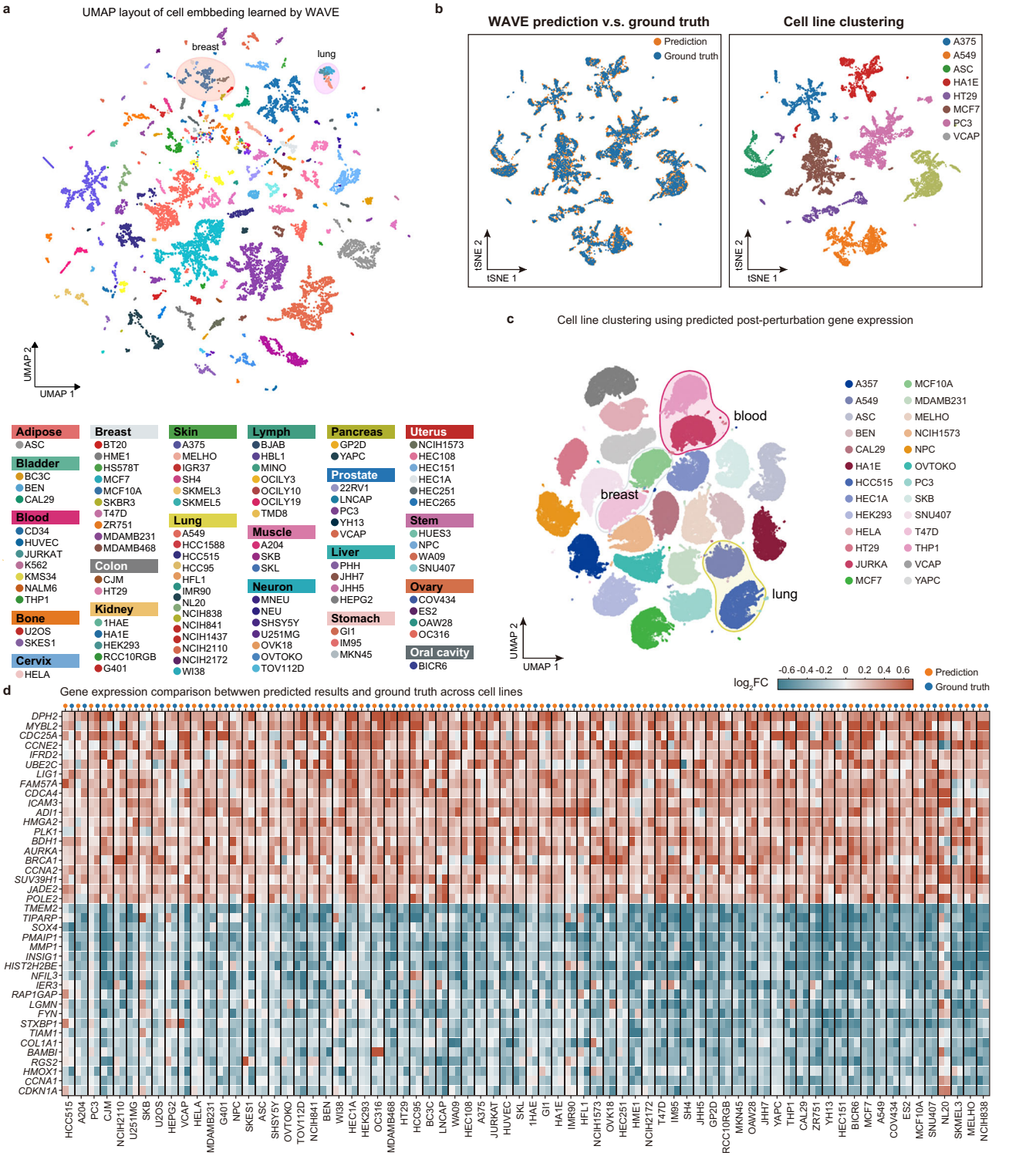


Fig. 3 | WAVE accurately reconstructs global perturbation effects and fine-grained transcriptional signatures across diverse LINCS cell lines. a UMAP visualization of cell lines based on their embeddings from the LINCS dataset. Cell lines are grouped and annotated according to their tissue of origin, with major groups (breast and lung) highlighted. **b** UMAP plots comparing true (left) and WAVE-predicted (right) perturbation expression profiles across eight major cell lines (A375, A549, ASC, HA1E, HT29, MCF7, PC3, VCAP). Each point represents a perturbation sample. **c** UMAP projection of predicted post-perturbation gene expression profiles for all cell lines, showing distinct clustering according to tissue type. **d** Heatmaps showing the predicted (left) and true (right) $\log_2$FC for the top 40 significantly altered genes across cell lines treated with entinostat. The overall similarity between predicted and true profiles illustrates WAVE's ability to capture key transcriptional responses at the gene level.

and further reinforce its potential in drug repurposing and target discovery. The consistent biological relevance of these results emphasizes the utility of WAVE in capturing detailed transcriptional responses and elucidating the MoAs for a diverse range of therapeutic agents.

WAVE captures cell type–specific and mechanism-relevant transcriptional responses at single-cell resolution

Although bulk transcriptomic data provide approximations of average drug responses, they fail to capture the cellular heterogeneity underlying differential drug sensitivity. With the emergence of single-cell perturbation datasets such as sci-Plex, training predictive models directly on single-cell data becomes increasingly important. We curated a subset of the sci-Plex dataset by selecting cells treated with compounds at 10 μM for 24 h and their corresponding DMSO controls. This subset contained 110,640 single-cell profiles from three cell lines (MCF7, A549, and K562) perturbed by 141 compounds.

We trained WAVE on this dataset to predict post-perturbation gene expression at the single-cell level. A UMAP projection based on the predicted expression profiles revealed clear separation among the three cell lines (Fig. 4a), indicating that WAVE effectively preserves intercellular heterogeneity. Evaluation on the held-out test set showed strong agreement between predicted and true expression profiles, achieving a PCC of 0.87, an R^2 of 0.76, and a DGE PCC of 0.82 (Fig. 4b, c). These results confirm WAVE's robust predictive performance.

To further assess biological relevance, we visualized UMAP embeddings of the predicted expression profiles for A549 and K562, colored by functional pathway categories of compounds. WAVE's predictions recapitulated the high-dimensional structure of the true data, faithfully preserving compound-induced transcriptional programs (Fig. 4d). Notably, drugs targeting epigenetic regulation formed distinct clusters, consistent with the original sci-Plex findings[28], suggesting WAVE effectively captures the underlying MoAs of perturbations.

WAVE also accurately recovered known gene signatures for representative compounds. For instance, in the A549 cell line treated with 2-methoxyestradiol and barasertib, WAVE highlighted key genes such as *ITPRIPL1* and *SAXO5* involved in apoptosis and cytoskeletal organization[29,30], and *TMC1*, *KBTBD12*, and *AKAP10*, which are associated with mitotic regulation[31,32] (Fig. 4e). For roxadustat, WAVE predicted upregulation of *AKAP10*, consistent with hypoxia-inducible factor pathway activation[33]. These results demonstrate that WAVE captures not only expression-level concordance but also mechanistic insights from single-cell data. In addition, Supplementary Fig. 3 showed the predictive performance of WAVE for the top 10 genes under treatment with these four compounds in the MCF7 and K562 cell lines. It is worth noting that the model's predictions may exhibit slightly lower variance than raw data, acting as a denoiser to prioritize robust biological trends over technical noise inherent in single-cell RNA sequencing.

Together, these results highlight WAVE's capacity to predict accurate and biologically meaningful transcriptional responses to drug perturbations at the single-cell level.

WAVE-based virtual screening for LUAD drug discovery

We utilized WAVE to predict perturbational gene expression profiles in the A549 cell line, enabling a virtual screening approach for potential compounds that may reverse LUAD-associated transcriptional profiles. We used the TargetMol L6000 compound library, consisting of 4566 chemical compounds, as the dataset for virtual screening.

To characterize the transcriptional landscape of LUAD, we performed differential gene expression analysis using TCGA-LUAD data[34], by using the 978 LINCS landmark genes. The volcano plot revealed a set of significantly up-regulated and down-regulated genes in tumor samples compared to adjacent normal tissues (Fig. 5a). Among the most up-regulated genes, *PYCR1*, *TOP2A*, *KIF20A*, *EZH2*, and *BIRC5* are known to be associated with tumor proliferation, chromatin remodeling, and cell cycle regulation. Notably, PYCR1, involved in proline biosynthesis and metabolic adaptation, has been implicated in lung cancer progression and poor prognosis through its role in redox homeostasis and mitochondrial function[35]. TOP2A, a DNA topoisomerase essential for DNA replication, and EZH2, a histone methyltransferase, are well-established drivers of oncogenesis in LUAD and are associated with resistance to chemotherapy and poor patient outcomes[36,37].

The GSEA result revealed that up-regulated genes in LUAD were significantly enriched in pathways related to cell cycle progression, including chromosome segregation, mitotic cell cycle process, nuclear division, and regulation of mitotic cell cycle phase transition (Fig. 5b). This is consistent with the well-established role of cell cycle dysregulation as a hallmark of lung cancer and suggests hyperproliferative transcriptional programs in LUAD tumor cells[38]. Conversely, suppressed pathways were associated with inflammatory response, transmembrane receptor signaling, and molecular transducer activity, indicating potential immune evasion and signaling reprogramming in LUAD tumors. The down-regulation of genes such as *ADRB2*, coding for a beta-adrenergic receptor linked to anti-tumor immune responses and cellular stress pathways[39], may further contribute to the tumor's ability to escape immune surveillance and sustain oncogenic signaling.

To prioritize candidate compounds, we calculated a connectivity score (CS) between the LUAD disease signature and each WAVE-predicted drug-induced expression profile. The connectivity score quantifies the extent to which a compound-induced transcriptional program is similar to or opposite to a query disease signature[4]. The score ranges from −1 to +1, with strongly negative values indicating that the compound is predicted to reverse the up- and down-regulated pattern observed in LUAD, whereas positive values indicate concordance with the disease state. Based on this metric, we ranked compounds according to their predicted ability to counteract the LUAD transcriptional phenotype.

Subsequently, we computed the connectivity scores of TargetMol L6000 compounds against LUAD by using the predicted perturbed gene expression. Importantly, these compounds were identified solely through WAVE-predicted perturbational profiles, without prior knowledge of their clinical indications. This screening identified 283 compounds with negative connectivity scores, suggesting potential reversal effects against the LUAD disease signature and highlighting their possible therapeutic relevance (Fig. 5c). Figure 5d shows the top 50 compounds with the strongest predicted potential to reverse the LUAD-associated gene expression signature. Among them, Topotecan is an FDA-approved therapy for lung cancer, directly supporting the predictive power of WAVE. Several additional compounds are not yet approved for lung cancer but have been reported in preclinical or investigational studies to exert anti-cancer effects. For example, Trichostatin A, an HDAC inhibitor, has been shown to inhibit lung cancer cell proliferation by reactivating tumor suppressor genes through epigenetic regulation (Fig. 5c)[40]. Geldanamycin, an HSP90 inhibitor, destabilizes several oncogenic proteins and has demonstrated activity in EGFR-driven tumors (Fig. 5c)[41]. Homoharringtonine, approved for leukemia, has shown pro-apoptotic activity in lung cancer cells and may be a promising repurposing candidate. In addition, several untested compounds identified by WAVE, such as Ellipticine and polygalaxanthone III, have shown promising anti-cancer activity in preclinical studies. Ellipticine, a known topoisomerase II inhibitor, has been shown to exert potent anti-cancer effects in lung cancer models. Although not yet clinically approved for lung cancer, these compounds demonstrate the chemical diversity and therapeutic potential that can be explored using WAVE.

Additional compounds such as hydroxysafflor yellow A, have demonstrated various degrees of anti-inflammatory, antioxidant, or

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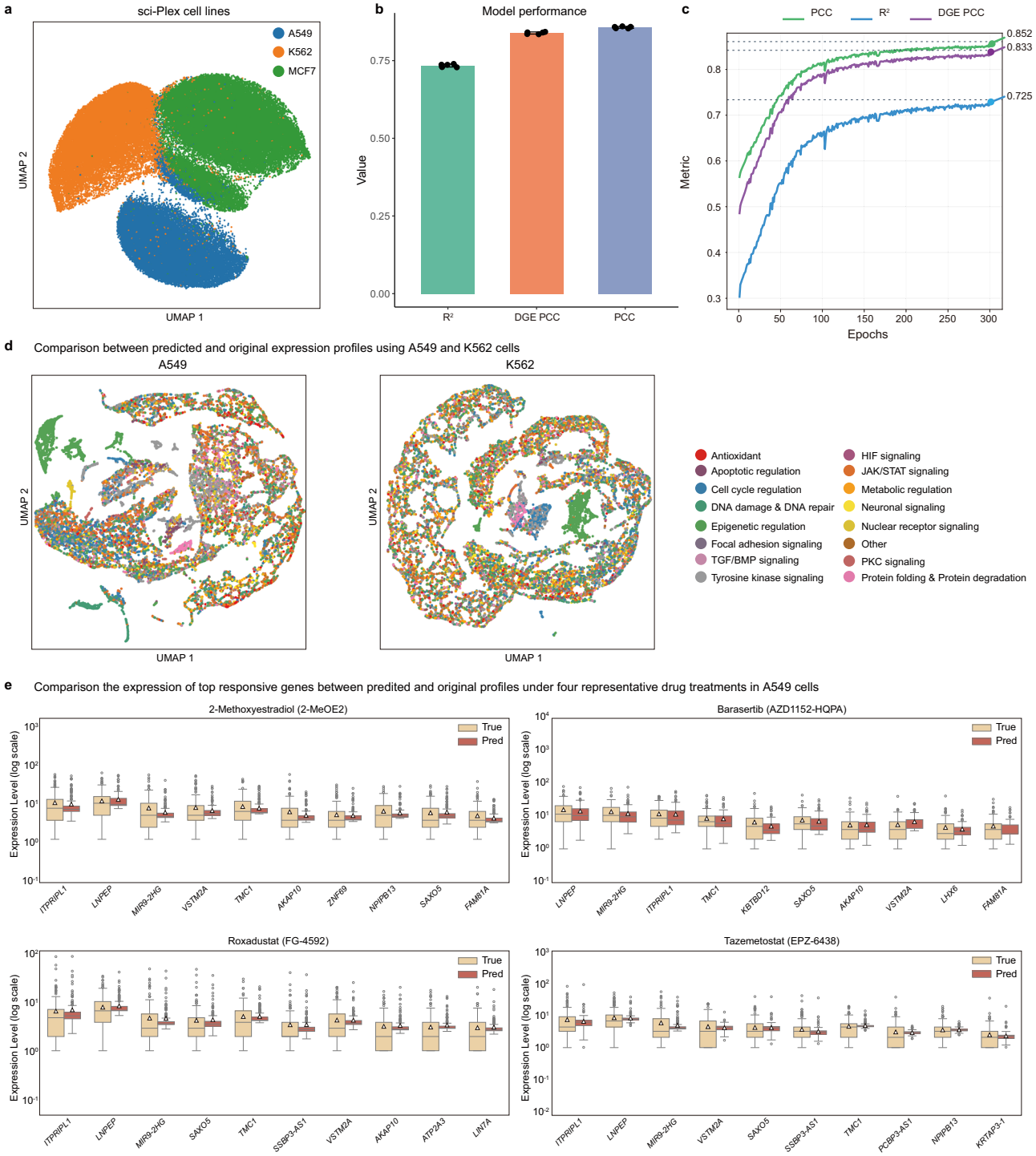


Fig. 4 | Performance of WAVE on the sci-Plex single-cell dataset. a UMAP visualization of sci-Plex profiles shows clear separation among the three cell lines (A549, K562, MCF7). **b** Model performance on the test set, evaluated using R^2, PCC, and DGE PCC across the full transcriptome. Bars show the mean performance across five cross-validation folds; error bars indicate s.d. across folds ($n = 5$). **c** Training curves showing the validation set performance over epochs for three metrics: PCC, R^2, and DGE PCC. **d** UMAP plots comparing predicted and true expression profiles for A549 and K562 cells, with points colored by functional pathway annotations. **e** Expression comparison of the top responsive genes under four representative drug treatments (2-methoxyestradiol, barasertib, roxadustat, and tazemetostat) in A549 cells. Boxplots compare predicted expression and true expression. The center line indicates the median, triangles indicate the mean, the box bounds represent the 25 and 75th percentiles (interquartile range), whiskers extend to 1.5 × the interquartile range, and points beyond the whiskers represent outliers. Each point represents an individual cell. Sample sizes were $n = 166$ cells (2-methoxyestradiol), $n = 115$ cells (barasertib), $n = 195$ cells (roxadustat) and $n = 203$ cells (tazemetostat).

anti-proliferative effects[42], though their precise anti-cancer mechanisms remain to be further elucidated. The presence of these agents in the top-ranking compounds suggests that WAVE may be capable of identifying not only known anti-tumor agents but also chemically diverse molecules with underexplored therapeutic potential.

To experimentally validate WAVE's computational predictions, we validated the top 283 candidate compounds prioritized by WAVE and

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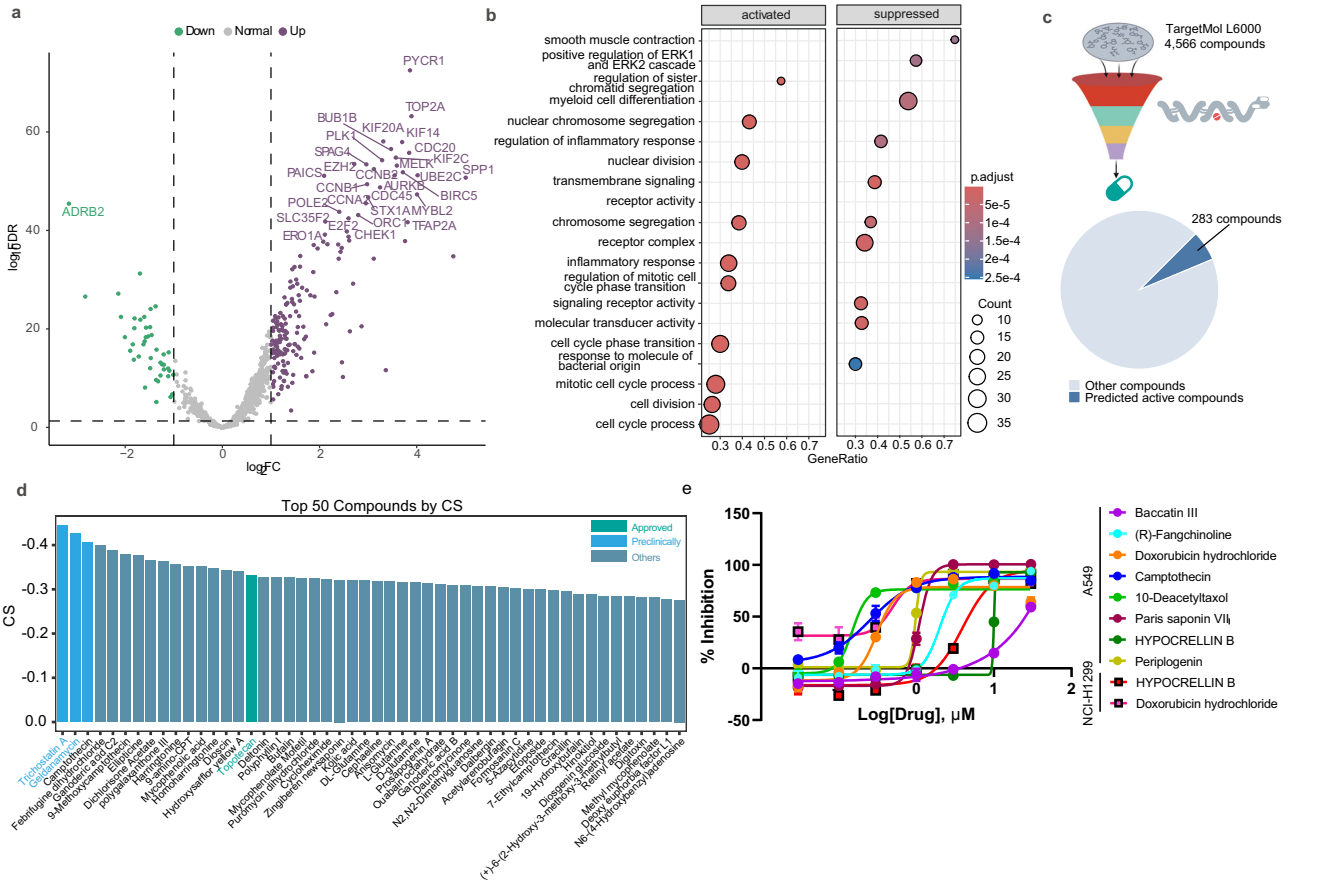


Fig. 5 | Prediction and experimental validation of drug-induced transcriptional reversal effects in LUAD. a Volcano plot of differentially expressed genes in LUAD versus normal tissues based on TCGA-LUAD data, using 978 LINCS 2020 landmark genes. **b** The GSEA result corresponding to LUAD versus normal tissue. Dot color indicates the adjusted *P* value (Benjamini–Hochberg FDR). **c** WAVE-predicted candidate compounds for LUAD. **d** Top-50 compounds predicted to have the strongest reversal effects on LUAD, based on the connectivity scores computed using the A549 cell line and a reference list of up-regulated and down-regulated genes. **e** Dose response curves of representative WAVE-nominated compounds in A549 and NCI-H1299 cells measured by CCK-8 viability assay after 48 h treatment. Compounds tested in A549 cells (Baccatin III, (R)-Fangchinoline, Doxorubicin hydrochloride, Camptothecin, 10-Deacetyltaxol, Paris saponin VII, HYPOCRELLIN B, Periplogenin) and in NCI-H1299 cells (HYPOCRELLIN B, Doxorubicin hydrochloride) are indicated in the figure legend. Data are shown as mean ± s.d.; $n = 3$ technical replicates per concentration.

performed a single-dose CCK-8 viability screen in two NSCLC cell lines, A549 and NCI-H1299. Under this condition, 10 compounds in A549 and 13 compounds in NCI-H1299 reduced cell viability to ≤50%, indicating that WAVE prioritization enriches for candidates with experimentally detectable anti-proliferative activity.

To focus follow-up experiments on the most potent candidates while accounting for cell line–specific sensitivity, we further defined a "high-effect" subset using stringent, cell line–specific cutoffs, defined as viability ≤35% in A549 cells and ≤70% in NCI-H1299 cells. Representative high-effect hits were selected for subsequent dose–response assays. The concentration–response curves of selected compounds generated in A549 and NCI-H1299 (Fig. 5e) exhibited obvious dose-dependent decreases in cell viability, supporting the quantitative nature of the observed effects.

To provide an orthogonal phenotypic readout beyond metabolic viability, we further assessed apoptosis by Annexin V/PI flow cytometry in A549 cells. Compared with vehicle controls (DMSO; Supplementary Fig. 4a), treatment with representative high-effect hits that were active in both cell lines (e.g., Baccatin III, Camptothecin, Paris saponin VII and Hypocrellin B) increased Annexin V-positive populations (Supplementary Fig. 4b-e), consistent with an apoptosis-associated phenotype. Together, these data support that WAVE-nominated compounds not only reduce viability but can also induce apoptotic responses in LUAD cell models. These compounds, identified purely through computational methods, could serve as valuable candidates for further experimental validation and ultimately contribute to the development of more effective treatments for LUAD.

Discussion

Predicting drug-induced transcriptional responses is a central challenge in modern computational pharmacology. While experimental datasets such as LINCS 2020 and sci-Plex have enabled large-scale exploration of compound-induced gene expression, they remain limited by cost, throughput, and biological context coverage. As a result, computational models have been developed to fill these gaps, enabling in silico prediction of transcriptomic perturbations. Nevertheless, many existing models face key limitations. For instance, DeepCOP and MultiDCP heavily rely on fixed feature representations and often struggle with generalization to untested compounds or cell types. Generative models such as CPA capture transcriptional variation to some extent, but they do not explicitly model the molecular structure, which may reduce their sensitivity to untested compounds.

WAVE addresses these limitations by introducing a generative framework based on β-VAE, which explicitly disentangles the reconstruction of baseline (unperturbed) gene expression from perturbation-induced transcriptional changes. By decomposing the prediction task into denoising baseline expression and modeling differential expression, WAVE enhances its sensitivity to compound-

induced biological effects. The fusion of cell-specific latent expression embeddings and molecular fingerprints enables WAVE to learn compound–cell interactions in a more specific manner.

Compared with previous models, WAVE demonstrates clear advantages in both accuracy and generalization. It consistently outperforms existing methods in predictions for unseen compounds and unseen cell types. Notably, WAVE achieves strong performance not only on bulk data from LINCS 2020 but also on single-cell perturbation data from sci-Plex, capturing cell-lineage-specific transcriptional heterogeneity and compound mechanism-related signatures. This broad applicability highlights WAVE's robustness and scalability across different experimental and biological contexts.

Recent work has also highlighted that conclusions regarding perturbation model performance can depend strongly on the calibration and biological relevance of the evaluation metrics used[43]. In line with this view, we assessed WAVE using multiple complementary metrics, including profile-level correlation, differential-expression-focused metrics, and ranking-based precision/recall measures. This multi-metric evaluation provides a more balanced assessment of model performance and helps avoid over-interpreting any single metric in isolation.

Moreover, WAVE's predictions are not limited to numerical metrics. Through downstream analyses such as GO and GSEA analyses, WAVE enables meaningful biological interpretation. In the context of lung cancer, WAVE successfully prioritized structurally diverse compounds with strong predicted connectivity to disease-specific expression signatures. This suggests that WAVE is not only capable of reproducing known effects but also of uncovering candidate therapeutic compounds, potentially expanding the chemical search space beyond conventional scaffold classes.

Despite these strengths, WAVE still has limitations. First, although WAVE models transcriptional responses at the gene level, it does not incorporate prior biological networks or pathway-level constraints, which might help in learning biologically coherent response patterns. Second, the current version of WAVE uses static control expression profiles and does not explicitly account for dose-response or time-series dynamics, which are known to influence drug effects. Incorporating such temporal or dosage information into the model may further enhance its biological resolution and predictive power. Lastly, while the model performs well on known compound–cell combinations, future work is needed to rigorously validate its predictive ability on rare cell types or less-characterized compounds using prospective experimental data.

In summary, WAVE offers a unified and high-performance framework for modeling drug-induced gene expression. Its ability to generalize across diverse compounds and cell contexts, coupled with accurate recovery of biological signals, makes it a promising tool for virtual drug screening, drug repurposing, and MoA studies. As large-scale perturbation data continue to grow, models like WAVE will play an increasingly central role in accelerating data-driven discovery in translational medicine and pharmacogenomics. In addition, although WAVE achieves strong predictive performance, the current framework does not explicitly provide mechanistic interpretability regarding how specific molecular substructures or latent features contribute to transcriptional responses. Future work could incorporate interpretable architectures or attribution-based analyses to improve the interpretability of the model.

Methods

The WAVE algorithm

The goal of WAVE is to predict transcriptional responses of cells affected by drug perturbations, using unperturbed gene expression and chemical features as inputs. Formally, we define a dataset $D = \{(x_i^u, s_i)\}_{i=1}^N$, where $x_i^u \in \mathbb{R}^g$ represents the unperturbed gene expression profile of sample i, and $s_i \in \mathbb{R}^{2048}$ is the Morgan fingerprint encoding the structure of the compound applied to sample i. The objective is to learn a function f that estimates the perturbed gene expression profile $x_i^p \in \mathbb{R}^n$ as:

$$\hat{x}_i^p = f(x_i^u, s_i) \tag{1}$$

WAVE approaches this prediction problem by modeling the perturbation effect as a combination of reconstructed unperturbed expression and an estimated drug-induced differential expression, which encourages the model to focus more on the genes that are actually affected by the perturbation.

$$\hat{x}_i^p = Decoder(z_i) + \triangle x_i \tag{2}$$

where z_i is the latent embedding of x_i^u, Decoder(z_i) represents the denoised baseline gene expression derived from a variational autoencoder, and $\triangle x_i$ is the predicted perturbation-specific change in gene expression, modeled by a fusion network that combines the latent representation of gene expression and the embedding of the compound fingerprint. WAVE consists of three major components:

Gene expression encoder–decoder (GeneVAE). The GeneVAE module is a VAE that learns a latent representation $z_i \in \mathbb{R}^{256}$ of unperturbed gene expression $x_i^u \in \mathbb{R}^n$. The encoder is a two-layer feedforward neural network. Latent variable z_i is sampled via the reparameterization trick:

$$z = \mu + \sigma \cdot \varepsilon,\ \varepsilon \sim \mathcal{N}(0, I) \tag{3}$$

The decoder is another two-layer neural network that reconstructs x^u from z.

Drug feature network. This module is a two-layer feedforward network that encodes the compound fingerprint $s \in \mathbb{R}^{2048}$ into a drug embedding $d \in \mathbb{R}^{256}$:

Gene-drug fusion module. This module predicts the change in gene expression $\triangle x \in \mathbb{R}^g$ perturbed by the drug. It concatenates the encoded gene latent vector $z \in \mathbb{R}^{256}$ and the drug embedding $d \in \mathbb{R}^{256}$: then passes the combined vector through a three-layer network. The final predicted gene expression is computed as:

$$\hat{x}^p = \hat{x}^u + \triangle x \tag{4}$$

Training objective

WAVE is trained end-to-end using a composite loss function:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \alpha \cdot \mathcal{L}_{\text{Pearson}} + \beta \cdot \mathcal{L}_{\text{KL}} \tag{5}$$

Mean squared error loss. Measures the reconstruction accuracy between predicted $\hat{x}^p$ and observed perturbed gene expression x^p:

$$\mathcal{L}_{MSE} = \text{MSE}(\hat{x}^p, x^p) \tag{6}$$

Pearson correlation loss. Encourages the predicted and true expression vectors to be linearly correlated:

$$\mathcal{L}_{Pearson} = 1 - \text{corr}(\hat{x}^p, x^p) \tag{7}$$

where the correlation is computed over samples as the average Pearson correlation between predicted and observed gene expression vectors.

KL divergence loss. Regularizes the latent variable distribution in the VAE:

$$\mathscr{L}_{KL} = -\frac{1}{N}\sum_{i} 0.5 \cdot \sum_{j}(1 + \log \sigma_{ij}^2 - \mu_{ij}^2 - \sigma_{ij}^2) \tag{8}$$

Training configuration

No extensive hyperparameter search was performed. Instead, a fixed training configuration was adopted across all experiments, using the Adam optimizer (learning rate = 0.001), a batch size of 32, and a maximum of 300 training epochs. The best-performing model checkpoint was selected based on the validation DGE PCC.

Model evaluation

To quantitatively assess model performance in predicting drug-induced gene expression profiles, we employed four metrics: PCC, R^2, DGE PCC, and Top-50 DGE PCC.

The PCC measures the linear correlation between the predicted expression vector $\hat{x} \in \mathbb{R}^n$ and the ground truth expression vector $x \in \mathbb{R}^n$, and is defined as:

$$PCC = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(\hat{x}_i - \bar{\hat{x}})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(\hat{x}_i - \bar{\hat{x}})^2}} \tag{9}$$

where $\bar{x}$ and $\bar{\hat{x}}$ are the means of the true and predicted expression values, respectively.

The R^2 evaluates the proportion of variance in the true expression values that is explained by the predictions:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(x_i - \hat{x}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x}_i)^2} \tag{10}$$

The top-50 DGE PCC focuses on the most strongly perturbed genes. For each sample, we ranked genes by their absolute differential expression values $|x_i - x_i^u|$, where x_i^u denotes the unperturbed (control) expression vector. We then selected the top 50 differential genes and computed the PCC between predicted and true expression values within this subset.

For the most biologically relevant changes, we focused on the top 100 up- and down-regulated genes, reporting positive precision@100, positive recall@100, negative precision@100, and negative recall@100.

These metrics jointly reflect the model's capacity to correctly identify genes that are significantly up- or down-regulated in response to compound treatment.

Connectivity score

The connectivity score is a metric derived from GSEA to quantify the similarity or antagonism between two transcriptional profiles. The score ranges from −1 to 1, with a value of 1 indicating complete similarity between the query and reference profiles, and −1 indicating a complete reversal.

We calculate enrichment scores for the up-regulated and down-regulated gene sets separately. Let n be the total number of genes in the ranked reference profile and t be the number of genes in the input gene set. For each gene j in the input set (ranked by position $V(j)$ in the reference list), we compute:

$$a = \max_{1 \le j \le t}\left(\frac{j}{t} - \frac{V(j)}{n}\right) \tag{11}$$

$$b = \max_{1 \le j \le t}\left(\frac{V(j)}{n} - \frac{j-1}{t}\right) \tag{12}$$

The enrichment score is then defined as:

$$ES = \begin{cases} a, \ if\, a > b \\ -b, \ if\, b > a \end{cases} \tag{13}$$

Finally, the connectivity score (CS) is computed by comparing the directionality of the two enrichment scores:

$$CS = \begin{cases} ES_{up} - ES_{down}, \ if\ sign(ES_{up}) \neq sign(ES_{down}) \\ 0, \ otherwise \end{cases} \tag{14}$$

LINCS 2020 dataset processing

We processed the LINCS 2020 level 3 dataset to obtain high-quality compound-induced transcriptional profiles across multiple cell lines, following the preprocessing pipeline described in the TranSiGen[16] framework. The dataset contains consensus signatures computed using the MODZ method from multiple replicates of the same perturbation condition. Only the landmark genes (978 genes) directly measured by the L1000 platform were retained for downstream analysis, excluding inferred genes to ensure measurement accuracy. We selected profiles that met the following criteria: (i) dose is 10 μM; (ii) time point is 24 h; (iii) number of replicates used to compute the consensus profile is ≥2. (iv) Only cell lines with a sufficient number of high-quality perturbation signatures (≥50 compounds) were retained to ensure reliable model training and evaluation. For each cell line, a DMSO control profile was generated by averaging the corresponding level 3 control samples.

sci-Plex dataset processing

We processed the sci-Plex single-cell transcriptomic dataset to construct matched perturbation-control gene expression pairs suitable for model training and evaluation. This dataset includes gene expression profiles of individual cells treated with various compounds under different conditions. To ensure consistency with the bulk-level LINCS data and to focus on acute transcriptional responses, we applied the following filtering and preprocessing steps. Only cells treated with compounds at a concentration of 10 μM for 24 h were retained for analysis. DMSO-treated cells were used as the control group. We computed the variance of gene expression across all cells and selected the top 3,000 most highly variable genes for downstream analysis. For each perturbation profile, a corresponding control profile was generated by randomly sampling DMSO-treated cells from the same cell line.

GSEA enrichment analysis

GSEA was performed using the clusterProfiler[44] package, which was also used for visualization of the results. Genes were ranked according to their log2 fold changes. Enrichment significance was evaluated using the permutation-based GSEA test, and P values were adjusted for multiple comparisons using the Benjamini–Hochberg false discovery rate procedure. GO biological process terms with adjusted $P < 0.05$ were considered statistically significant.

In silico screening of active compounds for LUAD

DEG signatures for lung cancer were obtained from the TCGA database[34] and used as reference gene lists for connectivity score analysis. The TargetMol L6000 compound library was used as the source of candidate compounds. The WAVE model was applied to predict the compound-induced expression profiles affecting A549 cell line (three biological replications). Connectivity scores for each compound were then computed by comparing the predicted expression profiles with the lung cancer DEG signature.

Experimental setting of validating screened compounds for LUAD

Compound. The 283 compounds screened against A549 and NCI-H1299 cell line were from TargetMol (Boston, MA, USA).

Cell culture. A549 (Jinyuan, JY135) and NCI-H1299 (Jinyuan, JY149) cell lines were purchased from Shanghai Jinyuan Biotechnology Co., Ltd. (Shanghai, China). A549 cells were cultured in Ham F-12K medium (Jinyuan, JY-H009) supplemented with 10% fetal bovine serum (FBS) (Yeasen, 40132ES76) and incubated at 37 °C under a humidified atmosphere containing 5% (v/v) CO_2. NCI-H1299 cells were cultured in RPMI-1640 medium (Jinyuan, JY-H003) supplemented with the same 10% FBS and maintained under the identical incubation conditions.

Cell viability assays (CCK-8). A549 and NCI-H1299 cells were seeded in 384-well plates (LABGIC, 11610) 1500 cells per well in 50 µL complete medium. The next day, compounds were added as 1 µL of pre-diluted working solutions to achieve the final concentrations. After 48 h of treatment, CCK-8 reagent (Share-bio, SB-CCK8) was added 5 µL per well and incubated for 1 h at 37 °C. Absorbance at 450 nm was measured using SpectraMax M5 (Molecular Devices), and viability was normalized to controls.

Apoptosis analysis by Annexin V/PI flow cytometry. A549 cells were seeded in 6-well plates (Corning, 3516) 1.5×10^5 cells per well and treated the next day with DMSO or compounds at 3 µM for 24 h. Both floating and adherent cells were collected, washed with PBS, and resuspended in Annexin V binding buffer (GlpBio, GK10037). Cells were stained with Annexin V-FITC and propidium iodide (PI) (GlpBio, GK10037) in the dark for 15 min at room temperature and analyzed by flow cytometer FACSCanto II (BD).

Statistics and reproducibility

Model performance was evaluated using 5-fold cross-validation, and performance metrics were calculated as described in the corresponding Methods sections. Sample sizes for computational analyses were determined by the available datasets. Cell viability was assessed in three technical replicates. No statistical method was used to predetermine sample size. No data were excluded from the analyses. Definitions of error bars, sample sizes and replicate types are provided in the corresponding figure legends.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

For this study, we employed transcriptomic profiles from both bulk and single-cell perturbation experiments, derived from the LINCS and sci-Plex3 collections. The LINCS Level 3 gene expression data were obtained from the CLUE data portal [https://clue.io/releases/data-dashboard], and the sci-Plex3 dataset was downloaded from the Gene Expression Omnibus under accession number GSM4150378. RNA-seq STAR-Counts data for the TCGA-LUAD project were obtained from the Genomic Data Commons [https://portal.gdc.cancer.gov/projects/TCGA-LUAD]. The preprocessed datasets used in this study are available at [https://figshare.com/articles/dataset/Pretrained_WAVE_Model_and_Datasets_for_Drug-Induced_Gene_Expression_Prediction_and_TCGA-Based_Drug_Screening/29650832]. No new omics datasets were generated in this study. Experimental validation data generated in this study, including IC50 and flow cytometry data, are provided in the Source Data file. Source data are provided with this paper.

Code availability

The code for model training and analysis is available at https://github.com/ZJUFanLab/WAVE and is distributed under the MIT License.

References

1. Vincent, F. et al. Phenotypic drug discovery: recent successes, lessons learned and new directions. *Nat. Rev. Drug Discov.* **21**, 899–914 (2022).
2. Chen, B. et al. Reversal of cancer gene expression correlates with drug efficacy and reveals therapeutic targets. *Nat. Commun.* **8**, 16022 (2017).
3. Du, P., Fan, R., Zhang, N., Wu, C. & Zhang, Y. Advances in integrated multi-omics analysis for drug-target identification. *Biomolecules* **14**, 692 (2024).
4. Lamb, J. et al. The connectivity map: using gene-expression signatures to connect small molecules, genes, and disease. *Science* **313**, 1929–1935 (2006).
5. Yang, W. et al. Genomics of drug sensitivity in cancer (GDSC): a resource for therapeutic biomarker discovery in cancer cells. *Nucleic Acids Res.* **41**, D955–D961 (2012).
6. Nusinow, D. P. et al. Quantitative proteomics of the cancer cell line encyclopedia. *Cell* **180**, 387–402.e16 (2020).
7. Subramanian, A. et al. A next generation connectivity map: L1000 platform and the first 1,000,000. *Profiles Cell* **171**, 1437–1452.e17 (2017).
8. Woo, G. et al. DeepCOP: deep learning-based approach to predict gene regulating effects of small molecules. *Bioinformatics* **36**, 813–818 (2020).
9. Zhu, J. et al. Prediction of drug efficacy from transcriptional profiles with deep learning. *Nat. Biotechnol.* **39**, 1444–1452 (2021).
10. Pham, T.-H. et al. Chemical-induced gene expression ranking and its application to pancreatic cancer drug repurposing. *Patterns* **3**, 100441 (2022).
11. Pham, T.-H., Qiu, Y., Zeng, J., Xie, L. & Zhang, P. A deep learning framework for high-throughput mechanism-driven phenotype compound screening and its application to COVID-19 drug repurposing. *Nat. Mach. Intell.* **3**, 247–257 (2021).
12. Wu, Y., Liu, Q., Qiu, Y. & Xie, L. Deep learning prediction of chemical-induced dose-dependent and context-specific multiplex phenotype responses and its application to personalized Alzheimer's disease drug repurposing. *PLoS Comput Biol.* **18**, e1010367 (2022).
13. Mering, C. V. STRING: a database of predicted functional associations between proteins. *Nucleic Acids Res.* **31**, 258–261 (2003).
14. Lotfollahi, M. et al. Predicting cellular responses to complex perturbations in high-throughput screens. *Mol. Syst. Biol.* **19**, e11517 (2023).
15. Hetzel, L. et al. Predicting cellular responses to novel drug perturbations at a single-cell resolution. (NeurIPS, 2022).
16. Tong, X. et al. Deep representation learning of chemical-induced transcriptional profile for phenotype-based drug discovery. *Nat. Commun.* **15**, 5378 (2024).
17. Adduri, A. K. et al. *Predicting cellular responses to perturbation across diverse contexts with State* Cell, 52. https://doi.org/10.1101/2025.06.26.661135 (2025).
18. Zeng, X. et al. Deep generative molecular design reshapes drug discovery. *Cell Rep. Med.* **3**, 100794 (2022).
19. Higgins, I. et al. -VAE: learning basic visual concepts with a constrained variantional framework. (ICLR, 2017).
20. David, L., Thakkar, A., Mercado, R. & Engkvist, O. Molecular representations in AI-driven drug discovery: a review and practical guide. *J. Cheminform.* **12**, 56 (2020).
21. Panapitiya, G. et al. Impact of molecular representations on deep learning model comparisons in drug response predictions. Preprint at https://doi.org/10.26434/chemrxiv-2024-g95j5-v2 (2024).

22. Shanmugam, G., Rakshit, S. & Sarkar, K. HDAC inhibitors: targets for tumor therapy, immune modulation and lung diseases. *Transl. Oncol.* **16**, 101312 (2022).
23. Wiltshire, T. et al. BRCA1 contributes to cell cycle arrest and chemoresistance in response to the anticancer agent irofulven. *Mol. Pharmacol.* **71**, 1051–1060 (2007).
24. Wang, P. et al. IFI27 enhances bladder cancer immunotherapy response by modulating regulatory T cell enrichment. *J. Cancer* **15**, 6616–6630 (2024).
25. Ginn, L. et al. A TIAM1-TRIM28 complex mediates epigenetic silencing of protocadherins to promote migration of lung cancer cells. *Proc. Natl. Acad. Sci. USA* **120**, e2300489120 (2023).
26. Lyu, S. et al. Fyn knockdown inhibits migration and invasion in cholangiocarcinoma through the activated AMPK/mTOR signaling pathway. *Oncol. Lett.* **15**, 2085–2090 (2017).
27. Rao, J. & Pulukuri, S. Matrix metalloproteinase-1 promotes prostate tumor growth and metastasis. *Int J. Oncol.* **32**, 757–765 (2008).
28. Srivatsan, S. R. et al. Massively multiplex chemical transcriptomics at single-cell resolution. *Science* **367**, 45–51 (2020).
29. Deng, S. et al. ITPRIPL1 binds CD3ε to impede T cell activation and enable tumor immune evasion. *Cell* **187**, 2305–2323.e33 (2024).
30. Chen, E. et al. A novel hepatocellular carcinoma-specific mTORC1-related signature for anticipating prognosis and immunotherapy. *Aging* **15**, 7933–7955 (2023).
31. Xu, K., Yang, L., Zhao, D., Wu, Y. & Qi, H. AKAP3 synthesis is mediated by RNA binding proteins and PKA signaling during mouse spermiogenesis. *Biol. Reprod.* **90**, 1–14 (2014).
32. Ballesteros, A. & Swartz, K. J. Regulation of membrane homeostasis by TMC1 mechanoelectrical transduction channels is essential for hearing. *Sci. Adv.* **8**, 5550 (2022).
33. Zhao, R. et al. Metabolites and gene expression in the myocardium of fasting rats in an acute hypoxic environment. *BMC Genomics* **24**, 251 (2023).
34. TheCancer Genome Atlas Research Network et al. The Cancer Genome Atlas Pan-Cancer analysis project. *Nat. Genet.* **45**, 1113–1120 (2013).
35. Westbrook, R. L. et al. Proline synthesis through PYCR1 is required to support cancer cell proliferation and survival in oxygen-limiting conditions. *Cell Rep.* **38**, 110320 (2022).
36. Kou, F. et al. TOP2A promotes lung adenocarcinoma cells' malignant progression and predicts poor prognosis in lung adenocarcinoma. *J. Cancer* **11**, 2496–2508 (2020).
37. Xu, M. et al. Treating human cancer by targeting EZH2. *Genes Dis.* **12**, 101313 (2024).
38. Hanahan, D. & Weinberg, R. A. Hallmarks of cancer: the next generation. *Cell* **144**, 646–674 (2011).
39. Mohammadpour, H. et al. β2 adrenergic receptor–mediated signaling regulates the immunosuppressive potential of myeloid-derived suppressor cells. *J. Clin. Investig.* **129**, 5537–5552 (2019).
40. Chen, X. et al. The epigenetic modifier trichostatin A, a histone deacetylase inhibitor, suppresses proliferation and epithelial–mesenchymal transition of lens epithelial cells. *Cell Death Dis.* **4**, e884–e884 (2013).
41. Abdullah, O. & Omran, Z. Geldanamycins: potent Hsp90 Inhibitors with Significant Potential in Cancer Therapy. *IJMS* **25**, 11293 (2024).
42. Jiang, M., Zhou, L., Xu, N. & An, Q. Hydroxysafflor yellow A inhibited lipopolysaccharide-induced non-small cell lung cancer cell proliferation, migration, and invasion by suppressing the PI3K/AKT/mTOR and ERK/MAPK signaling pathways. *Thorac. Cancer* **10**, 1319–1333 (2019).
43. Miller, H. E., Mejia, G. M., Leblanc, F. J. A., Wang, B. & Swain, B. *Deep Learn.-based Genet. perturbation models do outperform uninformative baselines well-calibrated Metr.* https://doi.org/10.48550/arXiv.2204.13545 (2025).
44. Yu, G., Wang, L.-G., Han, Y. & He, Q.-Y. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS J. Integr. Biol.* **16**, 284–287 (2012).

Author contributions

X.F. and J.L. conceived the study. T.L. and B.C. wrote and revised the manuscript, and led all computational work. X.D. and M.G. performed experimental validation during the revision process. B.Z., M.S., Q.Z., M.G., and X.D. participated in the discussions and provided valuable suggestions. All authors read and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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